Power Converter Analysis and Design

ASSIGNMENT - 1

1. Consider the periodic pulse waveform $v_{in}(t)$ shown in Fig. 1, which represents a unipolar rectangular pulse train with amplitude V_{in}, period T_s, and duty ratio D $(0 < D < 1)$. The waveform assumes a value of V_{in} for the interval $0 \leq t < DT_s$ and is zero for the remainder of the switching period. This type of excitation is commonly encountered in pulse-width modulated (PWM) power electronic circuits.

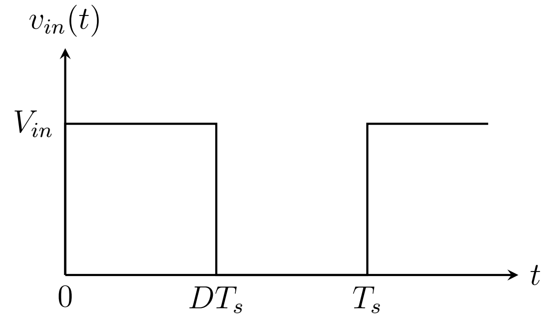


Figure 1: Rectangular pulse train with amplitude V_{in}, duty ratio D, and period T_s.

   (a) Derive the Fourier series representation of $v_{in}(t)$ in terms of the fundamental switching frequency $f_s = 1/T_s$ and its harmonics. Express the result in trigonometric form.

   (b) Identify the DC component, the amplitude of the fundamental component, and the general expression for the n^{th} harmonic.

   (c) Discuss the dependence of the harmonic spectrum on the duty ratio D and highlight the conditions under which certain harmonics vanish.

2. Refer to the *BOOSTXL-BUCKCONV R2.1* schematic provided by Texas Instruments. The output stage of the buck converter incorporates a second order LC low pass filter, which is realized by the series inductor L_1 and a parallel connected capacitor bank composed of five elements (C_4, C_5, C_6, C_7, and C_8). Capacitors C_4, C_5, C_6 are ceramic, while C_7 and C_8 are aluminum electrolytic. The combination of ceramic and electrolytic capacitors is commonly used in power converter designs to leverage the high frequency performance of ceramic capacitors and the high capacitance energy storage capability of electrolytic capacitors. The parallel arrangement of ceramic capacitors lowers the overall equivalent series resistance (ESR) near the switching frequency, thereby improving attenuation of switching frequency ripple. In contrast, electrolytic capacitors enhance converter transient performance under large

load variations. The resulting network not only reduces output voltage ripple and switching noise, but also enhances the transient response of the converter under dynamic load conditions. The simplified equivalent circuit of the output stage of buck converter is shown in Fig. 2.

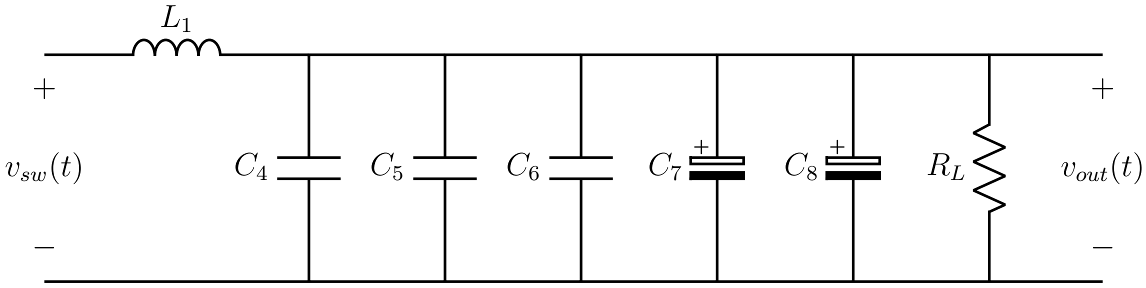


Figure 2: Simplified output stage of BOOSTXL-BUCKCONV R2.1 board.

(a) Determine the total equivalent capacitance of the parallel combination of capacitors C_4, C_5, C_6, C_7, and C_8, using the values specified in the TI schematic. Neglect parasitics and layout effects.

(b) Derive the transfer function $\dfrac{V_{\text{out}}(s)}{V_{\text{sw}}(s)}$ of the output stage.

(c) Determine the resonance frequency for $R_L \in [1\,\Omega,\ 10\,\Omega]$.

(d) Generate Bode plots (magnitude and phase) in MATLAB for $1\,\text{Hz} \leq f \leq 30\,\text{MHz}$ for $R_L \in [1\,\Omega,\ 10\,\Omega]$.

(e) Determine the attenuation (in dB) provided by the LC filter at the converter switching frequency (200 kHz).

Note: The schematic of the *BOOSTXL-BUCKCONV R2.1* board is provided at the end of this document.

3. Power electronic converters, while offering high efficiency and compact design, generate significant switching harmonics due to the fast transitions of semiconductor devices. These harmonics manifest in both the input and output waveforms, contributing to conducted electromagnetic interference (EMI). To comply with electromagnetic compatibility (EMC) regulations (e.g., CISPR standards), it is common practice to employ input filters that suppress high frequency switching components flowing back into the source.

   The buck converter shown in Fig. 3, operating in open loop, employs both an input LC filter (L_1, C_1) and an output LC filter (L_2, C_2) feeding a resistive load R_L. The switch and diode are assumed to be ideal. **Given parameters:** $V_{in} = 48\,\text{V}, \quad V_o = 36\,\text{V}, \quad f_s = 100\,\text{kHz}, \quad R_L = 6\,\Omega$

   (a) Sketch the transistor current waveform $i_T(t)$ over one switching period.

   (b) Derive analytical expressions for the DC components of the capacitor voltages and inductor currents in both filters.

   (c) Derive analytical expressions for the peak-to-peak ripple magnitudes of the input filter inductor current and capacitor voltages

   (d) Design the output filter such that:

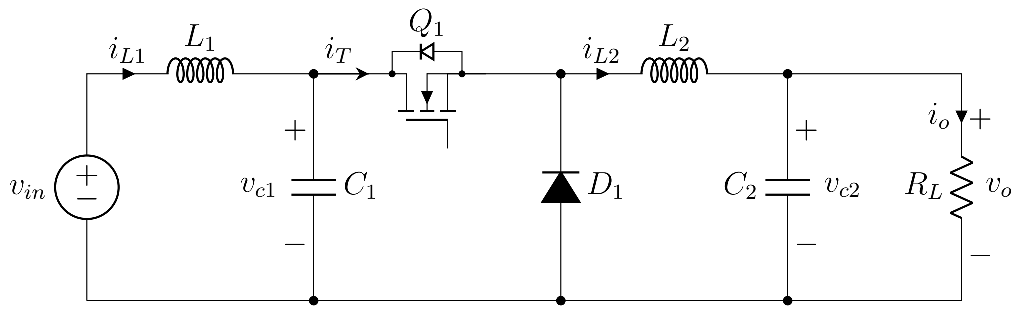


Figure 3: Buck converter with both input and output LC filters

- The peak-to-peak inductor current ripple satisfies $\Delta i_{L2} \leq 20\%\, I_o$.
- The peak-to-peak output voltage ripple satisfies $\Delta v_o \leq 2\%\, V_o$.

While designing L_2 and C_2, assume that the ripple in the input capacitor voltage v_{C1} is negligible.

(e) Select values for L_1 and C_1 such that:

- The peak-to-peak input filter capacitor voltage ripple satisfies $\Delta v_{C_1} \leq 4\%\, V_{C_1}$.
- The peak-to-peak input inductor current ripple satisfies $\Delta i_{L1} \leq 40$ mA.

While designing L_1 and C_1, assume that the ripple in the output inductor current i_{L2} is negligible.

4. Using the values of the filter components obtained in the previous question, perform the following simulation studies for the buck converter (shown in Fig. 4) in SIMULINK:

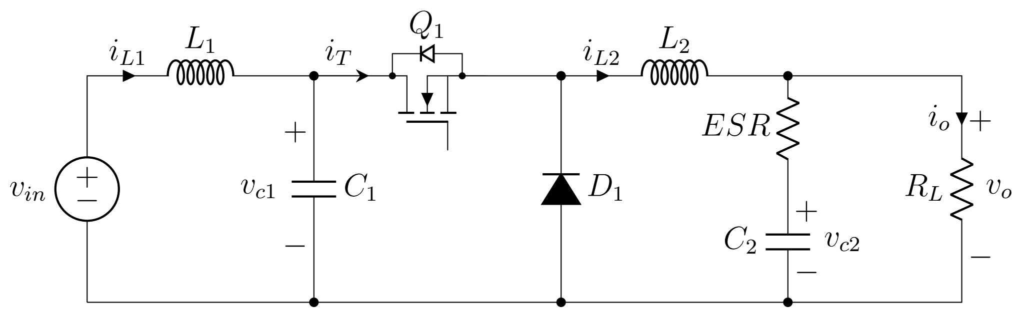


Figure 4: Buck converter with both input and output LC filters

(a) Simulate the converter with duty ratio of $D = 0.75$ and equivalent series resistance $ESR = 0\,\Omega$. Record the steady-state output voltage and the peak-to-peak ripple magnitude of the output voltage.

(b) Vary the ESR from 1 mΩ to 100 mΩ in at least ten uniform steps. For each case, determine the output voltage ripple and plot the peak-to-peak ripple voltage as a function of ESR. Assume, $R_L = 6\ \Omega$ and $D = 0.75$.

(c) Investigate the combined influence of load resistance and duty ratio on the output voltage with $ESR = 0\,\Omega$. Sweep the duty ratio from $D = 0.2$ to $D = 0.8$ in steps of 0.2. For each duty ratio, vary the load resistance from $R_L = 6\ \Omega$ to 600 Ω in at least ten steps. For all the cases, note down the average output voltage and generate a parametric plot of the average output voltage versus the load resistance for different duty ratios.

5. A boost converter shown in Fig. 5, operates at a switching frequency of 50 kHz and is supplied with an input voltage that varies between 5 V and 10 V. The output is regulated at 15 V using a closed loop control.

   (a) Find the duty cycle range. (Assume all components are ideal and continuous conduction mode (CCM) operation).

   (b) A load resistor (R_L) of $15\,\Omega$ is connected across the output. If the maximum allowable inductor current ripple is 20% of the average inductor current, estimate the value of the inductor to be used in the boost converter.

   (c) Estimate the output capacitor value if the output voltage ripple specification is 1% of the average output voltage.

   (d) Determine the root mean square (RMS) value of the current in diode (D) and MOSFET (Q)

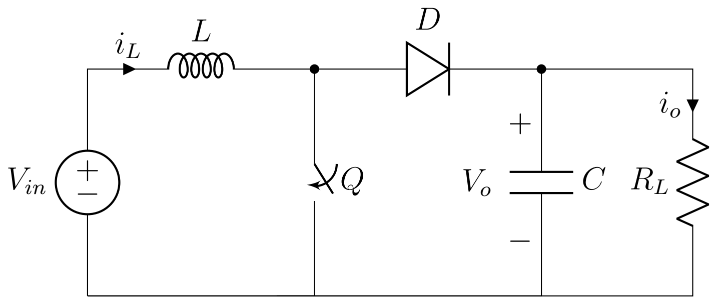
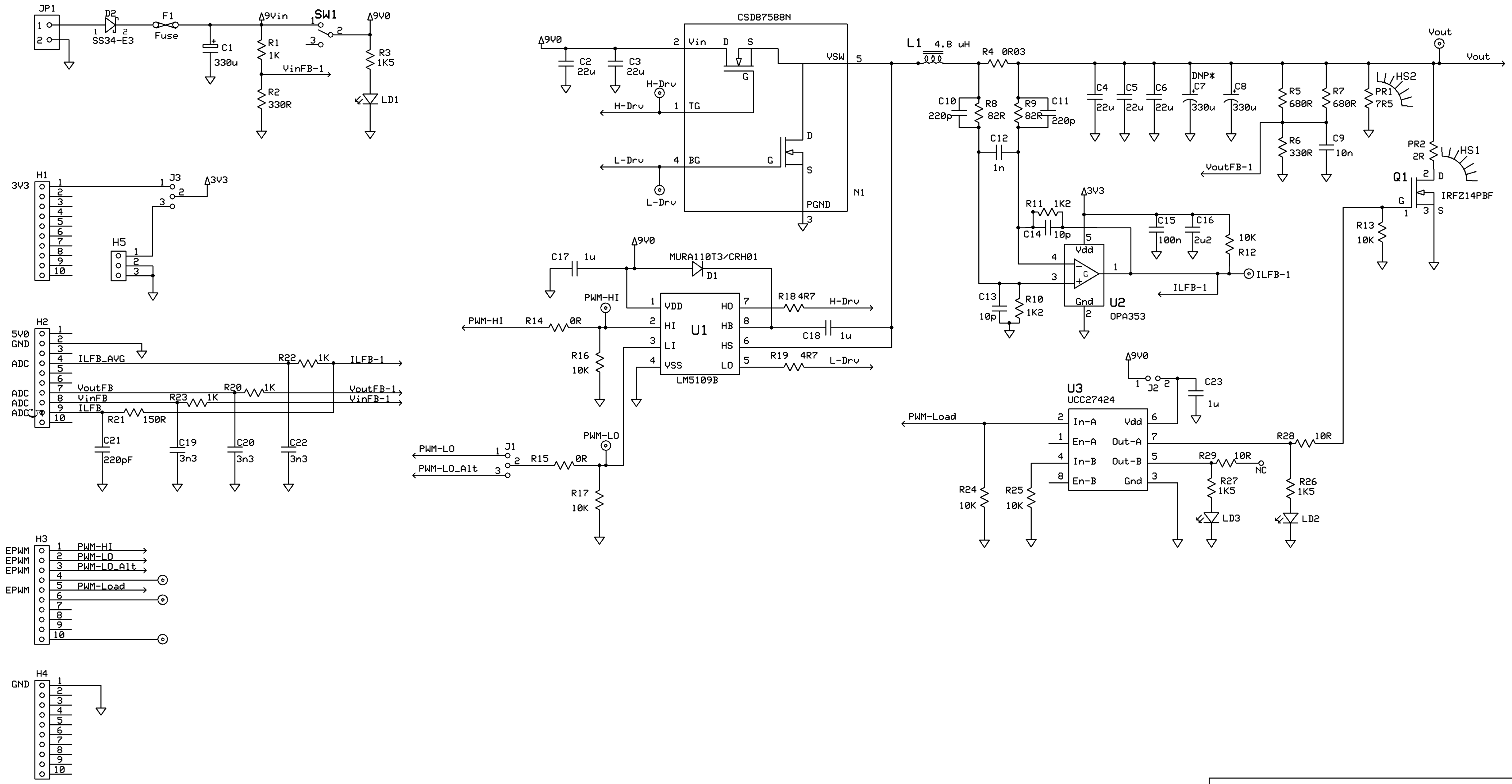


Figure 5: Boost Converter



DNP* = Do not populate

Texas Instruments		
C2000 - DPS BoosterPack Rev 2.1		
C2000 - HN	01/13/2015	Original Release
C2000 - TL	05/14/2020	Corrected known issues > C7, D1, R10, R11, U1-1 Readability edits > Net names, NC, U3